

Decision Making Under Uncertainty: 95-760

Recitation Session Week 5

PRACTICE PROBLEM FOR SUBMISSION

Instructions on how to submit: The problems can be handwritten or typed and should be uploaded to Gradescope by Saturday, September 26th at 11:59 pm. The problem will require you to modify the spreadsheet "Recitation W5.xlsx" available on Canvas. You should copy a screenshot of the spreadsheet into the PDF that you submit showing all the components of the problem (parameters, constraints, decision variables, and objective function). You don't need to submit the Excel document you use to solve the problems.

PROBLEM 1

You are very excited for next spring because you recently joined a gardening club! The club has asked you to help decide which foods to plant in its community garden.

The garden has 100 square feet of available planting space. You can choose from 10 different foods. Each food requires a different amount of space and has a different expected harvest.

To guarantee variety, the club wants to plant at least 4 different types of food. In addition, the club would like to grow some herbs. However, if an herb is planted, at least one fruit or vegetable must also be planted. If you decide to plant a particular food, you must plant at least the minimum quantity specified in the table below.

Food	Category	Space per Plant (sq. ft.)	Minimum Plants	Expected Harvest per Plant (lb.)
Tomato	Vegetable	4	4	10
Bell Pepper	Vegetable	2	6	5
Carrot	Vegetable	0.5	20	0.25
Lettuce	Vegetable	1	10	1.5
Strawberry	Fruit	1	12	1
Cucumber	Fruit	4	3	8
Zucchini	Vegetable	9	2	12
Basil	Herb	1	4	1
Cilantro	Herb	0.5	6	0.5
Parsley	Herb	0.5	4	0.6

Lastly, consider the following two requisites:

- The club wants to plant tomatoes, but tomatoes require companion plants. If tomatoes are planted, at least two types of herbs must be planted.
 - If more than 30 tomato plants are planted, at least 10 lettuce plants must also be planted.
 - The club wants to plant at least two of the following plants: tomato, bell pepper, and zucchini. However, if zucchini is planted, tomatoes cannot be planted.
- a) Formulate the problem as an integer programming model to help the harvesting club maximize the expected total harvest. Clearly define your decision variables, constraints, and objective function.
- OBJECTIVE FUNCTION: $\sum_{i \in F} e_i \cdot y_i$, where e_i is the expected harvest per plant.
- b) Solve the problem using Excel Solver.
- c) Present the results to the harvesting club.